03. Water Resources

01. Why do we consider water as a resource?

- Domestic purpose
- For organisms
- General hydroelectricity
- Helps in transport
- Industrial use

02. What is water scarcity?

It refers to the situation where people, unpolluted and fresh water is lower than the demand in a region.

03. What are the reasons for water scarcity?

- Over-consumption of groundwater for domestic purposes has depleted it.
- Surface water has been polluted by disposal of industrial waste and non-biodegradable substances.
- We are not allowing the filtration of rainwater to form groundwater by covering ground with concrete.
- Uneven distribution of rainfall.
- Over-irrigation has led to depletion of water.
- Industries over-exploit water.
- Unequal access to water between urban and rural population.
- Unequal distribution to water between farmers and industries.

04. What are the preventive measures of water scarcity?

- The judicious use of groundwater for domestic purpose.
- Industries should recycle and reuse water and we should not throw non-biodegradable substances.
- We have to keep our ground uncovered to allow rainwater to seep through the ground to groundwater.
- We have to grow trees to cool down the temperature.
- To avoid over-irrigation, dry crops should be grown instead of wet crop.
- During water crisis water should be equally distributed between urban and rural population.
- Water should be equally distributed between farmers and industries.
- Rainwater harvesting should be done.

05. What is dam?

A dam is a barrier across flowing water that obstructs, directs or retards the flow, often creating a reservoir, lake or impoundment.

06. Classify dams on the basis of their height and materials used for their construction.

Based on structure and the materials used, dams are classified as –

[i] Timber dams,
[ii] Embankment dams,
[iii] Masonry dams, with several subtypes.

According to height, dams can be categorized as –

[i] Large dams or alternatively as low dams,
[ii] Medium height dams,

[iii] High dams.

07. What are the advantages of dams?

- Hydroelectricity is generated.
- Water for irrigation is provided to the far-fetched fields through irrigation canals.
- It prevents occurrences of floods.
- It prevents scarcity of water by storing water in the reservoirs and canals
- Reservoirs act as a place of tourist attractions.
- Pisciculture is practiced in the reservoirs.
- The reservoirs are used for inland navigation.
- It provides water to the industries.

08. Mention the disadvantages of multi-purpose river-valley projects.

- People living on the banks of the rivers become homeless.
- Flora & fauna are destroyed on the banks of rivers.
- Farmers having fields on the banks of rivers lose their source of livelihood.
- Discrimination is done while supplying electricity. Hydroelectricity is mostly supplied to the industries and urban areas.
- Discrimination is done while supplying water. During dry seasons most of the water is supplied to the urban people and industries due to which farmers and rural people face water scarcity. Sometimes giving rise to riots.
- Dams have given rise to inter-state disputes and disputes between countries.
- The over-sedimentation of river bed in the upstream part has increased the occurrences of flood.
- The spawning of aquatic animals has become difficult due to prevalence of rocky bed in the downstream part of the river.
- Due to easy availability of water through the irrigation canals, farmers have changed the cropping pattern. They are growing wet crops instead of dry crops. This is leading to over-exploitation of water and is degrading the soil.
- The social gap between the big land holders and the small land holders has become wider.
- The stagnant water in reservoirs and irrigation canals is causing water borne diseases.
- Sometimes dams are triggering earthquakes.

09. Define the term rainwater harvesting.

Collecting and storing of rainwater to be used during dry months is called rainwater harvesting.

10. Mention the traditional methods of rainwater harvesting.

[i] Guls & Kuls.
[ii] Johads & Khadins.
[iii] Innundation canals.
[iv] Rooftop rainwater harvesting practiced in Rajasthan.

11. Describe how modern adaptations of traditional rainwater harvesting methods are being carried out to conserve and store water.

- Rooftop rainwater is collected using a PVC pipe.
- Filtered using sand and bricks.
- Underground pipe takes water to sump for immediate usage.

- Aayan

03. Water Resources

- Excess water from the sump is taken to the well.
- Water from the well recharges the underground.
- Take water from the well (later).

-THE END-

- Aayan